Nachiket Mahajan

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EDUCATION

***Bachelor of Technology in Electronics Engineering (*MIT AOE), Pune** ***(*2023 – Present)**

Specialization: Embedded Systems, PCB Design, Machine Learning, Computer vision

- **CGPA (Second year): 9.01**
- **SGPA (Second year): 9.32**
- **Modules:** Healthcare Technologies, Object-Oriented Programming (OOP) and Data Structures in C++, Computational Intelligence, Control System, ARM-Based Embedded Systems, Electronic Circuits and Applications

10th Grade, SSc **Jan 2009 - April 2021**

D.Y. Patil English medium School

- A Levels Equivalent: 84.20% in SSC.

12th Grade, HSc **Jan 2021 - April 2023**

D.Y Patil, Shahunagar

- First Class: 73.83%

WORK EXPERIENCE

R&D Intern | Embedded Systems Engineer **Jun 2025 - Present**

Inthink Technologies ***Remote, Pune***

- Designed and developed an **embedded healthcare system** for non-invasive diabetes monitoring.
- Developed an **IoT-based soil nutrient monitoring** system
- Worked on **CNN–based** human detection involving data processing, model implementation.
- Worked with STM32 microcontrollers (STM32L432KC, STM32WB55CGU6), embedded C firmware development and communication protocols including UART, I²C, RS-485, and BLE.

Data Automation Intern **Aug 2024- Jan 2025**

Zumax Innovation ***Remote, Pune***

- Developed a **Python-based** automation pipeline to process steel material data (thickness, length, weight, type) from CSV files, eliminating manual data entry and automatically logging date, time, and complete records.
- Built a web-based interface using Flask with HTML, CSS, JavaScript to manage and visualize the data.

COURSES & CERTIFICATIONS

- **Digital Twin** - MITAOE | ISA Pune Section | IEEE Bombay Section - *Certificate*
- **Iot Architectural Engineering** - L&T - *Certificate*
- **Python Programming** – CodSoft - *Certificate*

Projects

- **Applied Machine learning techniques for the classification of schizophrenia based on features extracted from multi-channel EEG signals:** link
  Major Project – under the Guidance of Dr. Vipin P. Yadav
  Implemented ML algorithms for region-wise EEG signals analysis using Double Banana Montage and extracted spectral, statistical, and entropy-based features from multi-channel EEG signals for schizophrenia classification. Comparative evaluation across multiple ML models and feature selection methods demonstrated consistently high classification performance across frontal, temporal, parietal, and occipital brain regions, accuracies ranging from 86.84% to 96.05%

- **An Integrated Vision and Geolocation mapping System for Sand Grain Size Analysis:** link
  Designed and implemented a vision-based system Using Raspberry Pi for camera control and GPS-based location tagging, with image processing algorithms like Watershed and segmentation for automated grain detection and dimensional analysis and a Flask-based web interface to visualize the locations, store, and manage analysis results.

- **Real-Time Vision-Based Hand Gesture Recognition for Multi-Degree-of-Freedom Robotic arm Control:**
  under the Guidance of Dr. Abhilasha Joshi
  Designed and implemented a hardware-integrated, real-time hand gesture recognition system for controlling a multi-DOF robotic arm. The system used computer vision and machine learning to extract and classify hand landmarks via MediaPipe, and interfaced the trained model with a servo-driven robotic arm through serial communication for real-time motion control

- **Vision-Based Numerical Extraction from Multi Seven-Segment Displays in CNC Machines using C++:**
  Implemented a complete image processing pipeline in C++ including grayscale conversion, adaptive thresholding, noise trimming, digit segmentation, and seven-segment pattern decoding. Optimized algorithms were designed to accurately detect and interpret digits arranged in a single row (e.g., 123, 456, 4789) while minimizing computational complexity and hardware requirements, making the system suitable for real-time industrial monitoring applications.

TECHNICAL SKILLS

- **Programming Languages**: Python (Libraries: NumPy, Pandas, Matplotlib, Paho, Flask, CV2, sklearn, tenserflow), C++, Embedded C, HTML, CSS, Java.
- **Computer Vision & Image Processing:** image segmentation, morphological operations, feature extraction, object detection, and geometric measurement from 2D images.
- **PCB Design & Prototyping:** Experienced in 2-layer PCB design and real-life PCB prototyping, with practical knowledge of circuit simulation and design tools including Multisim, Proteus, KiCad.
- **Communication protocols:** I²C, SPI, UART/USART, RS-485 (Modbus RTU), Bluetooth Low Energy (BLE), Serial Communication, HTTP/REST
- **Embedded Software & Microcontrollers:** Skilled in using STM32CubeIDE, STM32CubeProgrammer, STM32CubeStudio, STM32CubeMonitor Arduino IDE and Keil for embedded system development. Familiar with STM32F4, STM32N6, STM32L4, STM32WB series, Rasbery pi, esp32/esp8266, 8051, Arduino microcontrollers.
- Efficient in making **attractive & interactive PPT**, good in excel and word also proficient in using latex

SOFT SKILLS

- Effective Communication – Completed a Communication Skills course, developing strong abilities **in public speaking**, **active listening,** and **interpersonal communication** for clear and confident expression.
- Team Collaboration & Presentation – Gained hands-on experience through **group discussions, team presentations**, and **collaborative activities**, enhancing teamwork, critical thinking, and the ability to convey ideas effectively.
- Creative skills: Crochet, knitting, Sketching and Painting, Cooking and Baking.